CSE430 Finals

Name: Subhi Bhuiyan
ID: 17201116

Ans. 1.

$$x(n) = \begin{bmatrix} 1 & 2 & 5 & 3 \end{bmatrix}$$

$$N = 4$$

$$x(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 2 + 5 + 3 \\ 1 - 2j - 5 + 3j \\ 1 - 2 + 5 - 3 \\ 1 + 2j - 5 - 3j \end{bmatrix}$$

$$x(k) = \begin{bmatrix} 11 \\ -4 + j \\ 1 \\ -4 - j \end{bmatrix}$$

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8 point DIT FFT

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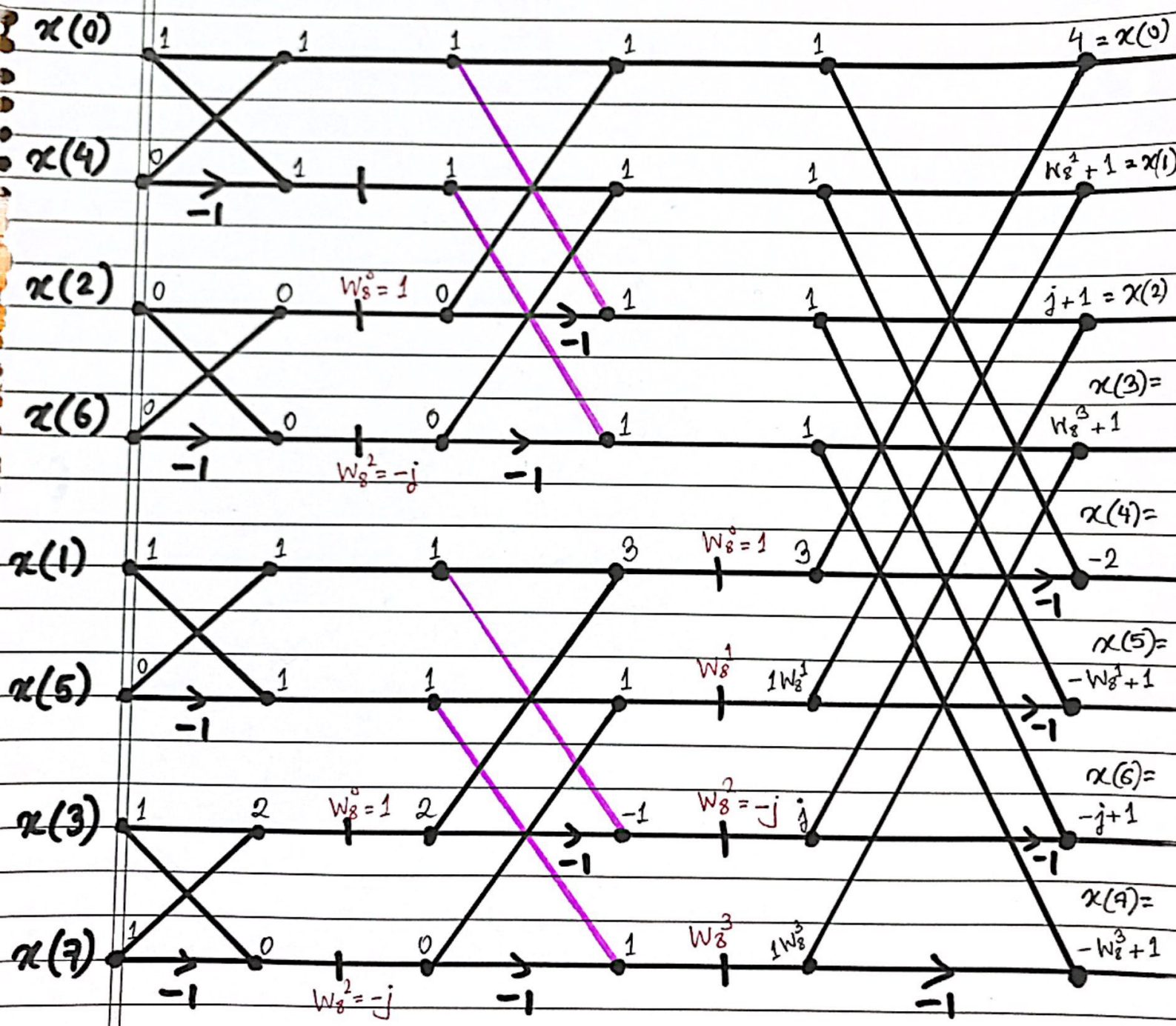
Ans 2:

$$x(n) = \{\overset{0}{1}, \overset{1}{1}, \overset{2}{0}, \overset{3}{1}, \overset{4}{0}, \overset{5}{0}, \overset{6}{0}, \overset{7}{1}\}$$



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$x(0) = 4$

$x(1) = W_8^1 + 1 = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j + 1 = \frac{2+\sqrt{2}}{2} - \frac{1}{\sqrt{2}}j$

$x(2) = 1 + j$

$x(3) = W_8^3 + 1 = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j + 1 = \frac{2-\sqrt{2}}{2} - \frac{1}{\sqrt{2}}j$

$x(4) = -2$

$x(5)\ 1 - W_8^1 = 1 - \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} = \frac{2-\sqrt{2}}{2} + \frac{1}{\sqrt{2}}j$

$x(6) = 1 - j$

$x(7) = -W_8^3 + 1 = \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} + 1 = \frac{2+\sqrt{2}}{2} + \frac{1}{\sqrt{2}}j$

$$x(k) = \left[4,\ \frac{2+\sqrt{2}}{2} - \frac{1}{\sqrt{2}}j,\ 1+j,\ \frac{2-\sqrt{2}}{2} - \frac{1}{\sqrt{2}}j,\ -2,\ \frac{2-\sqrt{2}}{2} + \frac{1}{\sqrt{2}}j,\ 1-j,\ \frac{2+\sqrt{2}}{2} + \frac{1}{\sqrt{2}}j \right]$$

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